

Rising Waters: A Machine Learning Approach to Flood Prediction

Project Executable Files

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This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	Yes
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

FloodPrediction/

```

|
|— app.py
|— requirements.txt
|— best_model.pkl
|— dataset.csv
|— templates/
|   |— index.html
|   |— result.html
|— static/
|   |— css/
|   |— images/
|— model_training.ipynb
|— README.md
|— screenshots/
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	IBM Cloud URL
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	IBM Cloud
Repository Link	https://github.com/dharaniseven/Flood-prediction-project
Demo Video Link	https://drive.google.com/file/d/17YHFCchbbk02E9yjGv72VWGVis_qEC2Cu/view?usp=drive_link

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

- 1.Install Python 3.8 or above.
- 2.Install required libraries using:
`pip install -r requirements.txt`
- 3.Open the project folder.
- 4.Run the Flask application:
`python app.py`
- 5.Open:
`http://127.0.0.1:5000`
- 6.Enter rainfall and weather parameters.
- 7.Click **Predict**.
- 8.View the flood prediction result.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction depends on dataset quality	Can be improved with larger datasets
2	Real-time weather API not integrated	Real-time weather API not integrated
3	Works with historical weather data only	Future Enhancement

